

Medvale Internal Medicine

Chapter 3H: Viral Hepatitis, Alcohol-Associated Liver Disease, MASLD/MASH, Autoimmune Hepatitis, and Drug-Induced Liver Injury

Original study-resource chapter. Coverage signals are drawn from the Medvale GI sequence and prior hepatology/biliary source material; copyrighted phrasing and images are not reproduced.

Why this chapter matters clinically

Hepatology problems often enter internal medicine through a narrow set of doors: an elevated aminotransferase panel, jaundice, new coagulopathy, incidental hepatic steatosis, alcohol-associated injury, positive viral hepatitis screening, or decompensated cirrhosis. The clinical task is to decide whether the patient has an acute hepatocellular injury, a chronic silent process with fibrosis risk, a cholestatic obstruction, or liver failure requiring urgent transfer. This chapter emphasizes pattern recognition and numeric anchors because the same symptoms - fatigue, nausea, anorexia, pruritus, jaundice, abdominal discomfort, and confusion - can represent anything from self-limited hepatitis A to fulminant acetaminophen toxicity.

A useful mental model is to separate liver disease into three overlapping questions. First, what is the biochemical pattern: hepatocellular, cholestatic, mixed, or impaired synthetic function? Second, what is the time course: acute days to weeks, subacute weeks to months, or chronic years? Third, is the patient compensated, decompensated, or failing? A patient with AST 900, ALT 1100, normal INR, and recent HAV exposure is different from a patient with AST 85, ALT 70, platelets 95, INR 1.6, ascites, and splenomegaly. The first has acute hepatocyte injury with preserved liver function; the second likely has advanced chronic liver disease even if the enzyme elevation is modest.

Graph-RAG focus: connect each disease to its trigger, lab pattern, diagnostic marker, complication, and management decision. For example, chronic HBV connects HBsAg persistence, HBV DNA activity, cirrhosis, HCC surveillance, and nucleos(t)ide analogue therapy; MASLD connects metabolic syndrome, steatosis, FIB-4, fibrosis, cardiovascular risk, and weight loss; acetaminophen toxicity connects dose/timing, glutathione depletion, massive aminotransferases, N-acetylcysteine, and acute liver failure.

Core illness scripts

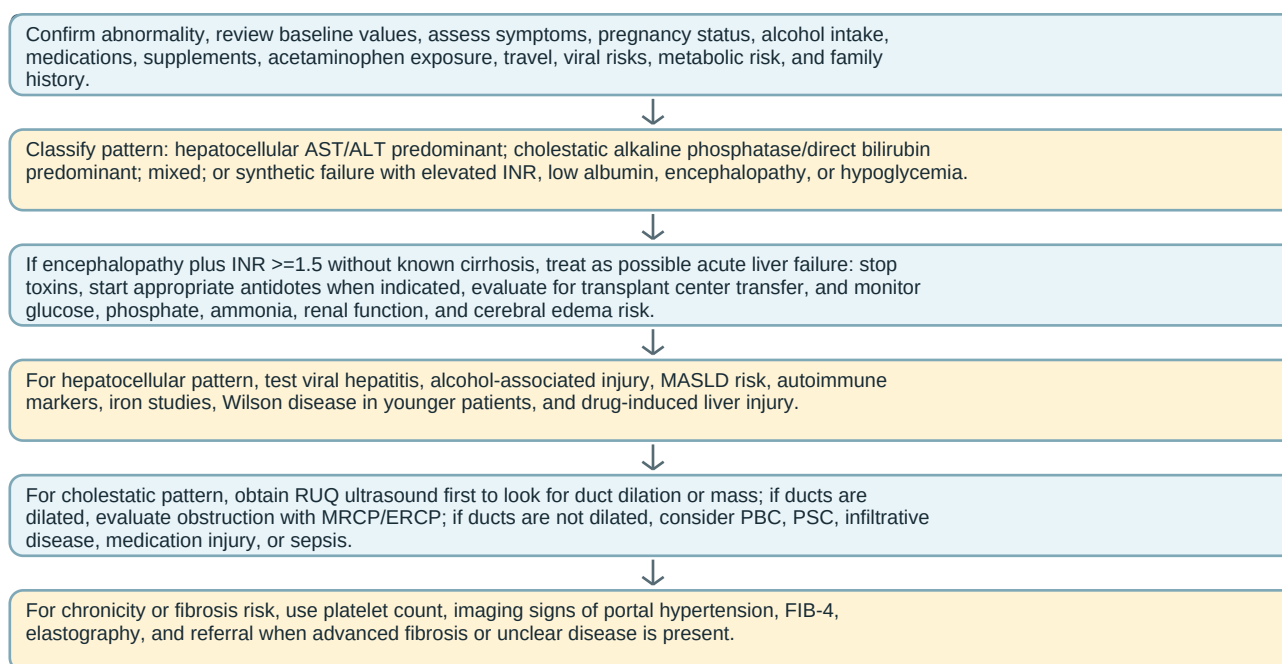
Syndrome	Typical signal	Key test pattern	Immediate clinical risk
Acute viral hepatitis	Malaise, anorexia, jaundice after exposure; sometimes fever	ALT/AST often hundreds to thousands; bilirubin variable; INR usually preserved unless severe	Acute liver failure if INR rises with encephalopathy
Chronic HBV/HCV	Often asymptomatic; abnormal ALT or positive screening	HBV: HBsAg/HBV DNA; HCV: anti-HCV then HCV RNA	Fibrosis, cirrhosis, hepatocellular carcinoma
Alcohol-associated hepatitis	Heavy alcohol use, jaundice, tender hepatomegaly, fever, leukocytosis	AST:ALT commonly >2:1 with absolute AST usually <300-500; bilirubin and INR may be high	Acute-on-chronic liver failure, infection, renal failure
MASLD/MASH	Obesity, T2DM, dyslipidemia, hypertension; often incidental steatosis	Mild ALT/AST elevation or normal enzymes; fibrosis risk assessed by FIB-4/VCTE	Advanced fibrosis, cirrhosis, HCC, cardiovascular events
Autoimmune hepatitis	Female predominance, other autoimmune disease, fatigue/jaundice	High IgG, ANA/ASMA; interface hepatitis on biopsy	Progression to cirrhosis; acute severe AIH

Syndrome	Typical signal	Key test pattern	Immediate clinical risk
Drug-induced liver injury	Temporal relationship to medication, supplement, toxin, or acetaminophen	Hepatocellular, cholestatic, or mixed pattern; acetaminophen causes very high AST/ALT	Acute liver failure; recurrent injury if drug is continued

Diagnostic approach to abnormal liver tests

The first step is not to memorize every rare etiology; it is to classify the pattern. Hepatocellular injury means AST and ALT are disproportionately elevated relative to alkaline phosphatase. Cholestatic injury means alkaline phosphatase and direct bilirubin are disproportionately elevated. Mixed patterns are common, especially with drug injury, biliary obstruction, and sepsis. Synthetic failure is identified by INR elevation, low albumin, thrombocytopenia from portal hypertension, hypoglycemia, or encephalopathy. Albumin changes slowly and is more helpful for chronic disease; INR can change rapidly and is a more urgent sign in acute liver injury.

Magnitude is helpful but not definitive. AST/ALT levels >1000 IU/L are classically associated with ischemic hepatitis, acetaminophen toxicity, and acute viral hepatitis, but severe autoimmune hepatitis, Budd-Chiari syndrome, and other toxins can also do this. Mild elevations can still hide advanced cirrhosis because fewer functioning hepatocytes remain to leak enzymes. Therefore, a normal or near-normal ALT does not exclude cirrhosis, and the platelet count, INR, bilirubin, albumin, imaging, and physical exam matter.



High-yield numeric anchors

- INR ≥ 1.5 plus any hepatic encephalopathy in a patient without established cirrhosis should trigger concern for acute liver failure.
- AST/ALT >1000 IU/L narrows the differential toward ischemic hepatitis, acetaminophen/toxin injury, and acute viral hepatitis, but does not exclude autoimmune or vascular causes.
- FIB-4 <1.3 is often used as a low-risk threshold for advanced fibrosis in many adult MASLD pathways; in adults older than 65, a higher cutoff is often used to avoid over-referral.

- HCC surveillance in cirrhosis is commonly performed every 6 months with ultrasound, with or without AFP depending on local practice.
- Severe alcohol-associated hepatitis is often assessed with Maddrey discriminant function, MELD, and dynamic response scores such as Lille after several days of therapy.

Viral hepatitis overview

Viral hepatitis is best learned by separating transmission, chronicity, diagnostic markers, and prevention. HAV and HEV are usually enterically transmitted and usually cause acute self-limited hepatitis. HBV, HCV, and HDV are blood/body-fluid transmitted and can become chronic. Chronicity is the major reason HBV and HCV matter for long-term internal medicine: patients may feel well for decades while fibrosis progresses, and both viruses can lead to cirrhosis and hepatocellular carcinoma.

Virus	Transmission	Chronic infection?	Diagnostic anchor	Prevention/treatment anchor
HAV	Fecal-oral; contaminated food/water; close contact	No chronic infection	IgM anti-HAV for acute infection	Vaccination; supportive care
HBV	Blood, sexual, perinatal, needle exposure	Yes; risk highest when acquired at birth/early childhood	HBsAg, anti-HBs, total anti-HBc, IgM anti-HBc, HBeAg, HBV DNA	Vaccination; tenofovir or entecavir for selected chronic HBV
HCV	Blood exposure; injection drug use; transfusion before screening era	Yes in many untreated patients	Anti-HCV screen followed by HCV RNA confirmation	Direct-acting antivirals; cure assessed by sustained virologic response
HDV	Requires HBV envelope; coinfection or superinfection	Can be chronic when HBV persists	Anti-HDV/HDV RNA in HBsAg-positive patients with risk or severe disease	Prevent with HBV vaccination; specialist treatment
HEV	Fecal-oral; undercooked pork/wild game in some regions; waterborne outbreaks	Usually no; chronic in some immunosuppressed patients	IgM anti-HEV or HEV RNA when suspected	Supportive; special concern in pregnancy and immunosuppression

Hepatitis A and E

Hepatitis A usually presents after an incubation period of several weeks with fatigue, anorexia, nausea, abdominal discomfort, dark urine, jaundice, and elevated aminotransferases. Adults are more likely than children to become symptomatic. It does not cause chronic hepatitis, so long-term management focuses on prevention and identifying contacts who need post-exposure prophylaxis. HAV vaccination is important for travelers, patients with chronic liver disease, men who have sex with men, people experiencing homelessness, and others with exposure risk.

Hepatitis E resembles HAV clinically but has several special associations. It is an important cause of waterborne hepatitis in endemic regions, can be acquired from undercooked pork or game in some settings, and can cause severe disease in pregnancy. In immunosuppressed patients, particularly solid-organ transplant recipients, HEV can become chronic. Because routine panels may not always include HEV, it becomes a targeted test when the story fits and the usual hepatitis workup is negative.

Hepatitis B

HBV is a DNA virus with a complex serology pattern that is high yield because each marker answers a different question. HBsAg means current infection. Anti-HBs means immunity, either from vaccination or recovery. Total anti-HBc means prior exposure to the viral core antigen and is not produced by vaccination alone. IgM anti-HBc suggests acute or recently acquired infection. HBeAg and HBV DNA reflect viral replication and infectivity, though management decisions also depend on ALT, fibrosis stage, age, pregnancy status, family history, and cirrhosis.

Pattern	HBsAg	Anti-HBs	Total anti-HBc	IgM anti-HBc	Interpretation
Susceptible	-	-	-	-	No infection and no immunity; vaccinate if eligible
Immune from vaccination	-	+	-	-	Vaccine response; no core antibody
Immune from past infection	-	+	+	-	Resolved natural infection; reactivation risk if severely immunosuppressed
Acute HBV	+	-	+	+	Acute infection; monitor severity and transmission precautions
Chronic HBV	+	-	+	-	Persistent infection; evaluate HBV DNA, ALT, fibrosis, HBeAg, and treatment need
Isolated core antibody	-	-	+	-	Possibilities include remote resolved infection, false positive, occult HBV, or window period if IgM positive

A common pitfall is to stop at a positive HBsAg. HBsAg tells you infection is present, but it does not tell you the phase of chronic HBV, fibrosis stage, or need for treatment. Patients with cirrhosis, high HBV DNA with active inflammation, or immunosuppression risk require careful evaluation. HBV also has an HCC risk even before cirrhosis in some groups, so surveillance decisions are not identical to HCV.

Hepatitis C

HCV is usually detected by antibody screening followed by confirmatory HCV RNA. A positive anti-HCV test alone indicates exposure, not necessarily active infection; spontaneous clearance or prior treatment can leave antibody positive and RNA negative. Active chronic HCV is defined by detectable RNA. The internal medicine task is to identify active infection, stage fibrosis, vaccinate against HAV/HBV when nonimmune, avoid excess alcohol, review drug interactions, treat with direct-acting antivirals when appropriate, and document cure with sustained virologic response.

Modern HCV therapy has changed exam logic. Genotype-specific regimens still exist, but simplified pangenotypic options are widely used for eligible treatment-naïve adults. Cure rates are high, and the major barriers are diagnosis, linkage to care, cirrhosis assessment, drug interactions, pregnancy considerations, renal/hepatic contraindications, and reinfection prevention. Patients with decompensated cirrhosis or prior treatment failure need specialist-guided regimens rather than routine simplified treatment.

Step	Clinical purpose	Practical anchor
Anti-HCV antibody	Screening for exposure	If negative and no recent exposure, active HCV unlikely; if positive, obtain RNA
HCV RNA	Confirms active viremia	Positive RNA = current infection; negative RNA = resolved or treated infection
Fibrosis staging	Determines cirrhosis status and HCC/variceal surveillance	Use labs, platelets, FIB-4, elastography, imaging, or biopsy when needed
Treatment	Eradicate infection and reduce fibrosis/HCC risk	Direct-acting antivirals; simplified regimens in eligible patients
SVR check	Documents cure	Undetectable RNA at least 12 weeks after treatment completion is sustained virologic response

Alcohol-associated liver disease

Alcohol-associated liver disease exists on a spectrum: steatosis, steatohepatitis, fibrosis, cirrhosis, and acute alcohol-associated hepatitis. Steatosis can occur rapidly and may improve with abstinence. Alcohol-associated hepatitis is a distinct inflammatory syndrome characterized by new jaundice, recent heavy alcohol exposure, systemic inflammation, and variable liver failure. Cirrhosis reflects chronic architectural remodeling and may remain clinically silent until ascites, variceal bleeding, encephalopathy, or jaundice appears.

The classic lab clue is AST greater than ALT, often with a ratio greater than 2, but absolute values are usually not in the thousands. AST and ALT commonly remain below 300 to 500 IU/L because alcohol injures mitochondria and because chronic disease reduces hepatocyte reserve. A very high AST/ALT should broaden the differential to ischemic injury, acetaminophen toxicity, acute viral hepatitis, autoimmune hepatitis, or vascular disease even if the patient drinks heavily.

Feature	Alcohol-associated steatosis	Alcohol-associated hepatitis	Alcohol-associated cirrhosis
Time course	Days to weeks of heavy use; often reversible	Acute/subacute jaundice on background of heavy use	Years of injury with fibrosis and portal hypertension
Common symptoms	Often asymptomatic; hepatomegaly	Jaundice, fever, RUQ pain, anorexia, malaise	Ascites, varices, encephalopathy, sarcopenia, splenomegaly
Labs	Mild AST/ALT; GGT may be elevated	AST:ALT often >2; bilirubin and INR may be high; leukocytosis	Platelets low, albumin low, INR high, bilirubin variable
Management center	Abstinence and metabolic/nutritional care	Alcohol cessation, nutrition, infection screen; corticosteroids in selected severe cases	Cirrhosis complication management, HCC surveillance, transplant evaluation

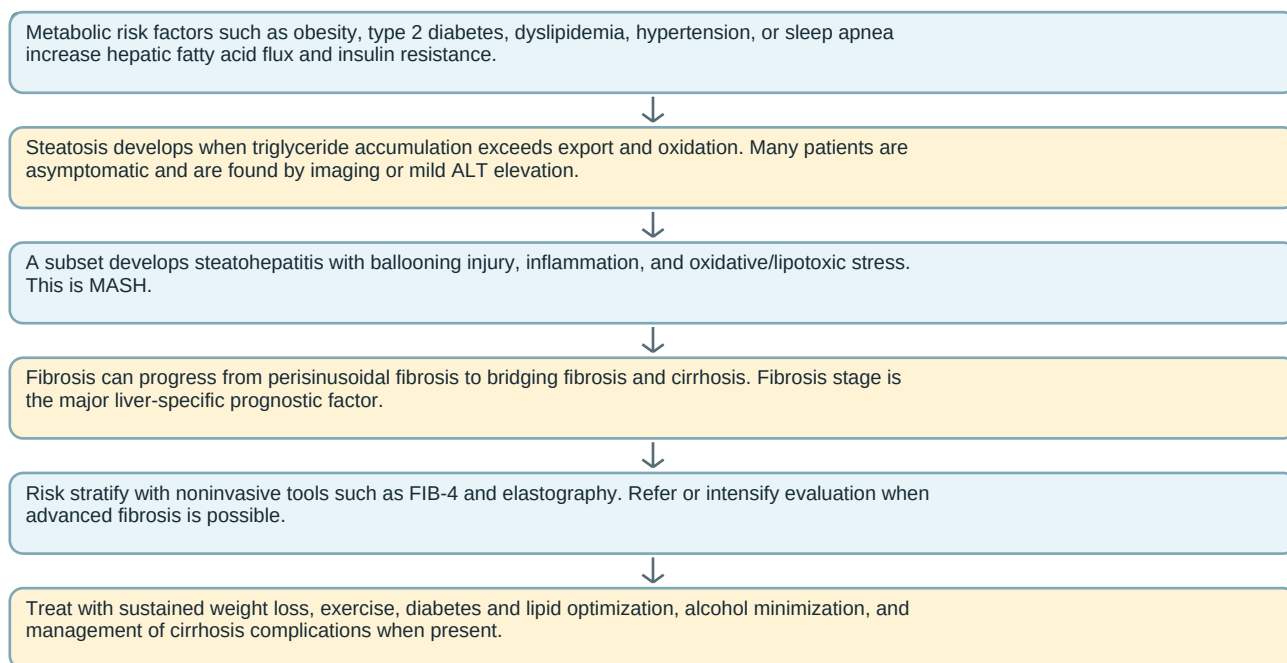
Severe alcohol-associated hepatitis is a high-mortality condition. Do not diagnose it by AST:ALT ratio alone. Look for recent heavy alcohol use, jaundice, elevated bilirubin, coagulopathy, systemic inflammation, and exclusion of biliary obstruction, viral hepatitis, acetaminophen toxicity, and infection. Corticosteroids are considered only after careful assessment for severe disease and contraindications such as uncontrolled infection, active GI bleeding, or renal failure. Nutrition, thiamine, infection evaluation, alcohol withdrawal management, and treatment of alcohol use disorder are core therapy, not extras.

Alcohol use disorder treatment belongs in the liver chapter because abstinence is disease-modifying. Brief counseling alone is usually insufficient. Pharmacotherapy may include agents such as acamprosate, naltrexone in selected patients without significant hepatic contraindications, gabapentin/topiramate in some contexts, or baclofen in advanced liver disease settings under clinician judgment. Medications must be individualized to renal function, liver function, opioid use, pregnancy, psychiatric comorbidity, and access to behavioral treatment.

MASLD and MASH

Metabolic dysfunction-associated steatotic liver disease (MASLD) refers to hepatic steatosis in a patient with cardiometabolic risk, after excluding other dominant causes. The older term NAFLD is still common in older literature. MASH corresponds to steatohepatitis: steatosis plus hepatocellular injury and inflammation, with or without fibrosis. Fibrosis stage is the main predictor of liver outcomes, but cardiovascular disease remains a leading cause of morbidity and mortality in this population.

MASLD is common because it tracks with insulin resistance, obesity, type 2 diabetes, dyslipidemia, hypertension, and sleep apnea. Aminotransferases may be normal, so a normal ALT does not exclude advanced fibrosis. This is important for graph RAG: do not link 'normal ALT' to 'no MASLD'; link metabolic risk to steatosis and fibrosis risk assessment. The management target is not simply to normalize enzymes; it is to reduce fibrosis risk, improve metabolic health, prevent cardiovascular events, and identify cirrhosis early.



Management target	Typical numeric/clinical anchor	Why it matters
Weight loss	3-5% may improve steatosis; larger sustained losses are usually needed for steatohepatitis/fibrosis improvement	Dose-response relationship between weight loss and liver histology
Exercise	Regular aerobic/resistance activity even without major weight loss	Improves insulin resistance and cardiometabolic risk
FIB-4	Low values help exclude advanced fibrosis; values ≥ 1.3 often trigger secondary assessment in many pathways	Cheap first pass using age, AST, ALT, platelets
Elastography	Liver stiffness estimates fibrosis; high values need specialist interpretation	Noninvasive fibrosis assessment
Diabetes therapy	GLP-1 receptor agonists and SGLT2 inhibitors may improve weight/metabolic risk; pioglitazone can help selected MASH patients but has side effects	Treats driver disease and cardiovascular risk

Pitfall: steatosis on ultrasound is not the same as harmless fatty liver. The central question is whether the patient has advanced fibrosis. Conversely, very mild steatosis does not explain every abnormal liver panel; viral hepatitis, alcohol, autoimmune disease, hemochromatosis, Wilson disease, biliary obstruction, and medication injury must be considered when the story does not fit.

Drug-induced liver injury and acetaminophen toxicity

Drug-induced liver injury (DILI) is a diagnosis of pattern recognition and exclusion. It can be hepatocellular, cholestatic, or mixed, and it can occur from prescription medications, over-the-counter products, herbal products, supplements, and toxins. The most important history elements are start date, stop date, dose changes, intermittent use, re-challenge, alcohol use, acetaminophen combination products, antibiotics, antiseizure drugs, statins, immune checkpoint inhibitors, anabolic steroids, and bodybuilding or weight-loss supplements.

Acetaminophen toxicity is special because there is an antidote. Toxic metabolism consumes glutathione and generates a reactive metabolite that injures hepatocytes. Early after ingestion, symptoms may be deceptively mild. Later, AST and ALT can rise into the thousands, INR increases, lactate may rise, renal injury may occur, and encephalopathy signals acute liver failure. N-acetylcysteine replenishes glutathione and is most effective early, but it is used even when timing is uncertain if acetaminophen level is detectable or liver injury is present.

DILI pattern	Lab pattern	Example associations	Clinical clue
Hepatocellular	ALT/AST predominant	Acetaminophen, isoniazid, some antivirals, immune checkpoint inhibitors	High ALT; may progress to acute liver failure if severe
Cholestatic	Alkaline phosphatase/direct bilirubin predominant	Amoxicillin-clavulanate, anabolic steroids, some antipsychotics	Pruritus, jaundice, dark urine, pale stools
Mixed	Both ALT/AST and alkaline phosphatase elevated	Many antibiotics, antiepileptics, herbal/supplement products	Pattern may evolve over time
Microvesicular/mitochondrial	Variable enzymes with systemic illness	Valproate, tetracycline, acute fatty liver of pregnancy	Lactic acidosis, hypoglycemia, encephalopathy can occur

Ask exact time, dose, formulation, co-ingestions, chronic use, alcohol use, suicide intent, and combination cold/opioid products.



Obtain acetaminophen level, AST/ALT, bilirubin, INR, creatinine, glucose, lactate, venous/arterial blood gas if ill, and pregnancy test when relevant.



If single acute ingestion with known time, plot level on Rumack-Matthew nomogram to determine treatment threshold.



Start N-acetylcysteine immediately if level is toxic, timing is unknown with detectable level, liver injury is present, or history suggests high-risk ingestion.



Monitor INR, mental status, glucose, acid-base status, renal function, phosphate, and transaminase trajectory; involve poison control and transplant center if acute liver failure develops.

Do not wait for a dramatic aminotransferase elevation before treating a high-risk acetaminophen exposure. The window for preventing injury is early, and the history may be unreliable. A low or undetectable level does not automatically exclude toxicity when ingestion was remote, repeated supratherapeutic dosing occurred, or delayed absorption is possible.

Autoimmune hepatitis

Autoimmune hepatitis (AIH) is an immune-mediated hepatocellular disease that can present as chronic mild aminotransferase elevation, acute hepatitis, or acute severe hepatitis. It is associated with other autoimmune diseases and classically has elevated IgG plus autoantibodies such as ANA, anti-smooth muscle antibody, or anti-LKM in selected phenotypes. Biopsy often shows interface hepatitis with plasma cell-rich inflammation, but the diagnosis integrates labs, serologies, histology, and exclusion of viral, drug, metabolic, and cholestatic diseases.

Treatment is immunosuppression. Prednisone or prednisolone, budesonide in selected noncirrhotic patients, and azathioprine are core options. The goal is biochemical remission, ideally normalization of ALT/AST and IgG. A common clinical error is to label a positive ANA as autoimmune hepatitis without IgG elevation, compatible histology, and exclusion of other causes; low-titer autoantibodies can appear in MASLD, viral hepatitis, alcohol-associated disease, and healthy individuals.

Clue	Supports AIH	Caution
Aminotransferases	Hepatocellular pattern, often fluctuating	Can mimic viral hepatitis or DILI
IgG	Typically elevated	Normal IgG does not fully exclude but makes classic disease less likely
Autoantibodies	ANA, ASMA, anti-LKM depending on type	Positive ANA alone is nonspecific
Biopsy	Interface hepatitis, plasma cells	Overlap syndromes and DILI can resemble AIH
Treatment response	Steroid responsiveness supports diagnosis	Relapse can occur if therapy is stopped too early

Ischemic hepatitis, Budd-Chiari syndrome, and vascular liver injury

Ischemic hepatitis, sometimes called shock liver, is a hepatocellular injury caused by hypoperfusion or hypoxemia. It usually occurs in the setting of shock, severe heart failure, respiratory failure, sepsis, arrhythmia, or massive bleeding. AST and ALT often rise dramatically, frequently into the thousands, and LDH may be strikingly elevated. The key management is correction of the underlying hemodynamic or oxygen-delivery problem; the liver injury often improves rapidly if perfusion is restored.

Budd-Chiari syndrome is hepatic venous outflow obstruction. It should be considered when abdominal pain, hepatomegaly, ascites, and liver injury occur in a patient with thrombophilia, myeloproliferative neoplasm, pregnancy/postpartum state, oral contraceptive exposure, malignancy, or inflammatory disease. Doppler ultrasound is often the initial test, and management may include anticoagulation, treatment of the underlying hypercoagulable state, angioplasty/stenting, TIPS, or transplant depending on severity.

Disorder	Mechanism	Clinical clues	Diagnostic anchor	Management anchor
Ischemic hepatitis	Low hepatic oxygen delivery	Shock, heart failure, hypoxemia, very high AST/ALT, high LDH	Clinical context plus rapid lab rise/fall	Restore perfusion/oxygenation; treat shock cause
Congestive hepatopathy	Chronic hepatic venous congestion	Right heart failure, hepatomegaly, ascites, mild cholestasis	Echo/right heart assessment; imaging congestion	Treat cardiac disease and volume status
Budd-Chiari	Hepatic vein outflow obstruction	Painful hepatomegaly, ascites, thrombophilia risks	Doppler ultrasound, CT/MRI venography	Anticoagulation; decompression procedures; transplant if severe
Portal vein thrombosis	Portal venous obstruction	Cirrhosis, malignancy, hypercoagulability, abdominal pain or variceal bleeding	Doppler ultrasound or CT/MRI portal venous phase	Anticoagulation selected by context; manage portal HTN

Metabolic and inherited liver disease links

Several inherited diseases appeared in earlier liver coverage and should be linked to abnormal liver tests, cirrhosis, and multisystem findings. Hemochromatosis causes iron overload; the classic triad is cirrhosis, diabetes, and skin hyperpigmentation, but many patients are identified earlier by elevated transferrin saturation and ferritin. Wilson disease causes impaired copper handling and should be considered in younger patients with hepatitis, neuropsychiatric symptoms, Coombs-negative hemolysis, low alkaline phosphatase in severe presentations, or unexplained liver disease. Alpha-1 antitrypsin deficiency can cause liver disease and panacinar emphysema.

Disease	Core defect	Liver clue	Extrahepatic clue	Test anchor
Hemochromatosis	Excess iron absorption/storage	Hepatomegaly, fibrosis, cirrhosis, HCC risk	Diabetes, cardiomyopathy, arthropathy, skin hyperpigmentation	Transferrin saturation, ferritin, HFE testing
Wilson disease	ATP7B copper transport defect	Hepatitis, cirrhosis, acute liver failure	Kayser-Fleischer rings, movement disorder, psychiatric symptoms, renal tubular issues	Ceruloplasmin, urinary copper, eye exam, liver copper
Alpha-1 antitrypsin deficiency	Misfolded A1AT accumulation and low protease inhibition	Neonatal hepatitis, chronic liver disease, cirrhosis	Early emphysema, panniculitis	A1AT level and phenotype/genotype
Celiac-associated liver tests	Immune/gluten-mediated disease	Mild transaminase elevation can occur	Diarrhea, iron deficiency, dermatitis herpetiformis	tTG-IgA plus total IgA

Integrated diagnostic tables

Presentation	Most important first distinctions	High-yield next tests	Red flags
Acute jaundice with AST/ALT high	Viral vs toxin/drug vs ischemic vs autoimmune	HAV IgM, HBV panel, HCV RNA if needed, acetaminophen level, INR, pregnancy test, autoimmune markers by context	INR ≥ 1.5 , encephalopathy, hypoglycemia, acidosis, renal failure
Chronic mild ALT elevation	MASLD/alcohol/viral/medications common; inherited/autoimmune less common but important	Hepatitis B/C screening, iron studies, metabolic assessment, ultrasound, FIB-4, autoimmune markers if unexplained	Platelets low, splenomegaly, nodular liver, ascites, family history
Cholestatic jaundice	Extrahepatic obstruction vs intrahepatic cholestasis	RUQ ultrasound, fractionated bilirubin, GGT, MRCP/ERCP if ducts dilated, AMA for PBC	Fever/hypotension/AMS suggests cholangitis; painless jaundice suggests malignancy
Positive hepatitis B screen	Active infection vs immunity vs resolved exposure	Interpret HBsAg, anti-HBs, anti-HBc; obtain HBV DNA/ALT/HBeAg if active	Pregnancy, immunosuppression, cirrhosis, high viral load, HCC risk
Steatosis on imaging	Benign steatosis vs fibrosis risk	FIB-4, platelets, metabolic panel, elastography when indicated	Diabetes, age, thrombocytopenia, AST > ALT, signs of portal HTN

Management principles

Management in hepatology is often staged. First, remove or treat the driver: stop a hepatotoxin, treat HCV, suppress HBV when indicated, stop alcohol, induce AIH remission, relieve obstruction, or improve metabolic risk. Second, stage fibrosis because fibrosis determines surveillance and prognosis. Third, prevent complications: vaccinate nonimmune chronic liver disease patients against HAV/HBV when appropriate, avoid unsafe alcohol and hepatotoxic drugs, screen for HCC when indicated, screen/manage varices in cirrhosis, and treat ascites or encephalopathy when present. Fourth, refer early when disease is advanced, diagnosis is unclear, acute liver failure is possible, or transplant evaluation may be needed.

Do not anchor on the enzyme level alone. The sickest liver patients can have falling AST/ALT because hepatocytes are dying or already depleted. Worsening INR, bilirubin, creatinine, lactate, hypoglycemia, phosphate abnormalities, and mental status are more concerning than a modest decrease in aminotransferases.

High-yield exam clues and common pitfalls

- HBsAg positive means current HBV infection; anti-HBs positive with anti-HBc negative means vaccine immunity.
- Anti-HCV positive requires HCV RNA to diagnose active infection.
- HAV and HEV usually do not become chronic; HBV, HCV, and HDV can.
- AST:ALT > 2 supports alcohol-associated injury, but values in the thousands should prompt a broader differential.
- MASLD/MASH can exist with normal aminotransferases; fibrosis risk stratification matters more than ALT alone.
- Acetaminophen toxicity may be clinically quiet early; treat high-risk cases with N-acetylcysteine while clarifying timing.
- Autoimmune hepatitis is not diagnosed by ANA alone; elevated IgG, compatible histology, and exclusion of mimics matter.
- Cholestatic jaundice should trigger ultrasound to look for obstruction before assuming hepatitis.
- Acute liver failure is a syndrome of liver injury plus coagulopathy plus encephalopathy, not simply a very high ALT.

Practice questions

Question 1. A 27-year-old health care worker has HBsAg negative, anti-HBs positive, and total anti-HBc negative. What does this pattern mean?

Explanation. This pattern indicates immunity due to vaccination. The lack of core antibody argues against natural infection. No active infection is present.

Question 2. A 58-year-old man with obesity, type 2 diabetes, and dyslipidemia has steatosis on ultrasound and ALT 38. What is the most important next clinical question?

Explanation. Whether he has advanced fibrosis. Normal or mild ALT does not exclude clinically important MASLD/MASH. Platelets, FIB-4, elastography, and signs of portal hypertension help risk-stratify.

Question 3. A patient presents with AST 4200, ALT 5100, INR 2.2, confusion, and uncertain ingestion history. What treatment should not be delayed?

Explanation. N-acetylcysteine should be started while acetaminophen level and other workup are pending, and the patient should be managed as possible acute liver failure with transplant-center consideration.

Question 4. A patient with chronic HCV has positive anti-HCV but undetectable HCV RNA. What does that mean?

Explanation. There is no evidence of current active viremia at that time. The patient may have cleared infection spontaneously or after treatment. Antibody can remain positive for life.

Question 5. A woman with fatigue, ALT 480, AST 390, high IgG, positive anti-smooth muscle antibody, and biopsy showing interface hepatitis most likely has what treatable condition?

Explanation. Autoimmune hepatitis. Treatment usually involves glucocorticoids with steroid-sparing immunosuppression such as azathioprine in appropriate patients.

Question 6. A patient with heavy alcohol use has jaundice, fever, leukocytosis, AST 190, ALT 70, bilirubin 16, and INR 2.0. What syndrome is suggested?

Explanation. Alcohol-associated hepatitis. Severity assessment, infection evaluation, nutrition, alcohol cessation treatment, and selective corticosteroid consideration are central.

Graph-RAG extraction map

Node	Core attributes to preserve	Important edges
HBV infection	Serologies: HBsAg, anti-HBs, anti-HBc, IgM anti-HBc; HBV DNA; HBeAg; chronicity; HCC risk	HBV -> cirrhosis; HBV -> HCC; vaccination -> anti-HBs; immunosuppression -> reactivation
HCV infection	Anti-HCV screen, HCV RNA confirmation, DAA treatment, SVR, fibrosis staging	HCV -> chronic hepatitis; HCV -> cirrhosis; DAA -> SVR; cirrhosis -> HCC surveillance
MASLD/MASH	Metabolic risk, steatosis, inflammation, fibrosis, FIB-4, elastography, weight loss	T2DM -> MASLD; MASH -> fibrosis; fibrosis -> cirrhosis; weight loss -> histologic improvement
Alcohol-associated hepatitis	Heavy alcohol, jaundice, AST:ALT >2, bilirubin/INR, Maddrey/MELD/Lille, steroids selected cases	Alcohol -> steatohepatitis; severe AH -> mortality; steroids -> response assessment
Acetaminophen toxicity	Dose/timing, Rumack-Matthew nomogram, N-acetylcysteine, acute liver failure	Acetaminophen -> glutathione depletion; NAC -> glutathione restoration; ALF -> transplant evaluation
Autoimmune hepatitis	IgG, ANA/ASMA, biopsy/interface hepatitis, prednisone/azathioprine	AIH -> chronic hepatitis; immunosuppression -> remission; untreated AIH -> cirrhosis
Abnormal liver tests	Hepatocellular vs cholestatic vs mixed vs synthetic failure	Pattern -> differential; cholestasis -> ultrasound; INR/encephalopathy -> ALF

Selected guideline anchors and sources used for clinical cross-checking

- CDC. Clinical Testing and Diagnosis for Hepatitis B. Updated 2025. Triple-panel screening with HBsAg, anti-HBs, and total anti-HBc.
- AASLD-IDSA HCV Guidance. Simplified HCV treatment algorithms and treatment-naïve adult guidance.
- Rinella ME et al. AASLD Practice Guidance on clinical assessment and management of NAFLD/MASLD, 2023.
- American College of Gastroenterology. Clinical Guideline: Alcohol-Associated Liver Disease, 2024.
- AASLD resources on autoimmune hepatitis, acute liver failure, and drug-induced liver injury, including N-acetylcysteine use.

This Medvale chapter is a study resource and not a substitute for local protocols, specialist consultation, or patient-specific medical decision-making.